

Marketing Experiment Sample Report: Coca-Cola's New Healthy Beverage Line

Introduction

This report aims to design a marketing experiment for Coca-Cola's new healthy beverage line, which includes low-calorie and organic options. The campaign will focus on a digital advertising initiative that highlights the health benefits and natural ingredients of the drinks. With increasing health awareness among consumers, this experiment will utilize a before-after design to assess the effectiveness of the digital advertising campaign. This design is selected to isolate the impact of the advertising without the complexity of multiple variable changes, allowing for clearer conclusions about causality.

Experiment Design

Independent and Dependent Variables

- **Independent Variable:** The introduction of a digital advertising campaign emphasizing health benefits.
- **Dependent Variable:** Sales of the new healthy beverage line.

Sample Size and Randomization

The experiment will be conducted in Austin, Texas, with Dallas, Texas serving as the control market. Both cities have similar demographics, with populations around 1 million. The entire populations will be exposed to the campaign, ensuring a large enough sample size for statistical significance.

Market

The selected markets are Austin (test market) and Dallas (control market), chosen for their comparable demographics and market conditions.

Timing and Duration

Sales data will be collected for both control and test markets over a three-month average from January to March. The experiment will run from April to June, allowing time to observe changes in sales due to the digital campaign.

Rationale for Choices

- A before-after design allows for clear measurement of the campaign's impact on sales.
- A large sample size ensures statistically significant results.
- Selecting similar markets minimizes confounding variables related to demographics and market conditions.
- The timing aligns with increased consumer interest in healthy products during spring and summer.

Causality Rules

The experiment adheres to the first three rules of causality:

1. **Temporal Precedence:** Sales data will be collected before and after the campaign.
2. **Covariation:** Changes in sales will be compared between control and test markets.
3. **No Alternative Explanations:** Randomization helps control for confounding variables.

However, it does not fully control for external factors that may influence results, such as competitive responses or economic changes.

Anticipated Issues

Potential issues include:

- **Seasonality:** Conducting both pre- and post-experiment sales during spring may not reflect performance in other seasons (e.g., winter).
- **External Factors:** New competitors entering the market or changes in consumer preferences could affect sales outcomes.

While these issues could impact results, the experiment will still provide valuable insights into the effectiveness of digital advertising on sales, helping to inform decisions about a broader marketing strategy.

Experiment 2.0: Social Media Advertising Version

An alternative version of the experiment can involve social media advertising instead of broader digital ads. This medium allows for better targeting and engagement with specific demographics.

a. Impact of Changing Medium

Switching to social media would enable more precise targeting based on user behavior and preferences, potentially leading to higher engagement rates compared to general digital ads.

b. More Informative Results

The social media version could yield more informative results due to:

- The ability to test multiple independent variables (e.g., different ad formats or messaging strategies).
- Enhanced analytics that provide deeper insights into consumer behavior and preferences.
- Generally lower costs and quicker implementation compared to traditional digital campaigns.

However, a limitation of social media advertising is that it may exclude segments of the audience who are less active online. In conclusion, both experiments offer valuable insights into marketing strategies for Coca-Cola's new healthy beverage line while addressing different aspects of audience engagement and response measurement.